higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N4

29 JULY 2019

This marking guideline consists of 10 pages.

Copyright reserved

QUESTION 1

1.1

$$f(x)=2x^{-4}$$
$$f(x+h)=2(x+h)^{-4}$$
$$¿\ 2\left[x^{-4}+(4)x^{-5}.h+(-4)(-5)x^{-6}h^2+\frac{(-4)(-5)(-6)x^{-7}h^3}{6}\right]$$
$$f(x+h)=2x^{-4}-8x^{-5}h+2x^{-6}h^2-40x^{-7}h^3$$
$$f(x+h)-f(x)=2x^{-4}-8x^{-5}h+2x^{-6}h^2-40x^{-7}h^3-2x^{-4}$$
$$f(x+h)-f(x)=-8x^{-5}h+2x^{-6}h^2-40x^{-7}h^3$$
$$\frac{f(x+h)-f(x)}{h}=\frac{-8x^{-5}h+2x^{-6}h^2-40x^{-7}h^3}{h}$$
$$¿\ -8x^{-5}+2x^{-6}h-40x^{-7}h^2$$
$$\lim_{\substack{h\to\theta\\h\to 0}}\left[\frac{f(x+h)-f(x)}{h}\right]=-8x^{-5}+2x^{-6}(0)-40x^{-7}(0)^2$$
$$¿\ -8x^{-5}$$

(4)

1.2

$$y=(5x+1)^{\frac{1}{2}}$$
$$let u=5x+1 \quad y=u^{\frac{1}{2}}$$
$$\frac{du}{dx}=5\quad\frac{dy}{du}=\frac{1}{2\sqrt{u}}$$
$$\frac{dy}{dx}=\frac{du}{dx}\times\frac{dy}{du}$$
$$¿\ 5\times\frac{1}{2\sqrt{5x+1}}$$
$$¿\ \frac{5}{2\sqrt{5x+1}}$$

(4)

1.3

$$\frac{dx}{dy}=\frac{1}{y}-0+2\pi\sin y-\frac{3}{4y}-\frac{1}{4\sqrt{y^3}}$$

(5)

1.4 $\frac{dy}{dx}=3x^2-2$

$3x^2=2$

$x^2=\pm\frac{2}{3}$

$x_1=0,816; x_2=-0,816$

$y_1=(0,816)^3-2(-0,816)-4$

$y_1=-5,089$

$y_2=(-0,816)^3-2(-0,816)-4$

$y_2=-2,911$

$\therefore$ Co-ordinates of the turning points are: (-0,816; -2,911) and (0,816; -5,089)

$P.O.I:$

$\frac{d^2y}{dx^2}=6x$

$x=0$

$y=(0)^3-2(0)-4$

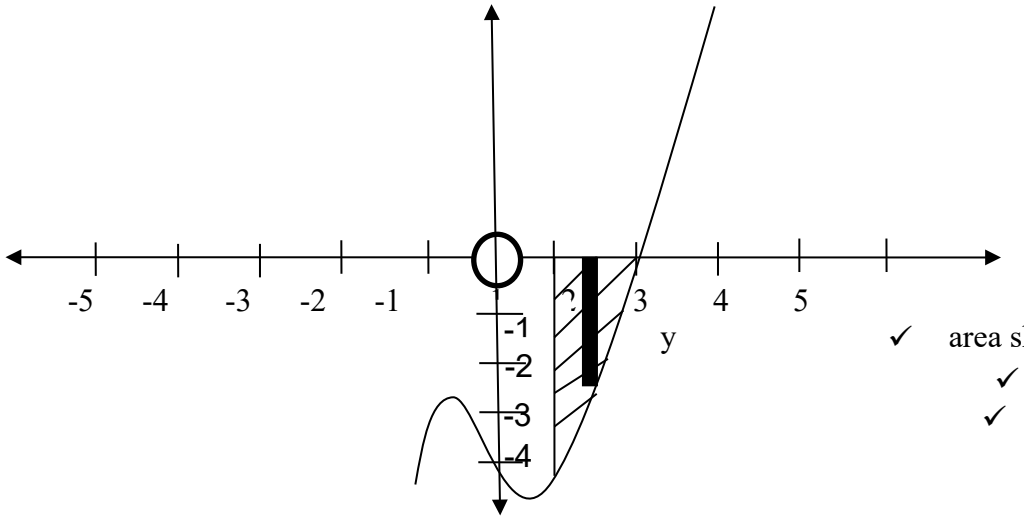
$y=-4$

$\therefore P.O.I.=(0;-4)$ (7)

[20]

QUESTION 2

2.1 2.1.1



✓ area shaded
✓ strip
✓ shape

(3)

2.1.2
$$Aox=\int_1^2 \left(x^3-2x-4\right)dx$$
$$¿\left[\frac{x^4}{4}-x^2-4x\right]_1^2$$
$$¿-8-(-4,75)$$
$$Aox=-3,25$$
$$¿3,25u^2$$
(4)

2.2 2.2.1
$$\int\left(\cos t-\sec^2 t-(2t)^{\frac{1}{2}}+2e^{-2t}+\frac{2}{t}-4^{4t}+4\right)dt$$
$$\sin t-\tan t-\frac{2}{3}(2t)^{\frac{3}{2}}-e^{-2t}+2\ln^t-\frac{4^{4t}}{4\ln^4}+4t+c$$
(7)

2.2.2
$$\left[\ln 3x-\ln x\right]_2^4$$
$$\left\{\left[\ln 3(4)-\ln 4\right]-\left[\ln 3(2)-\ln 2\right]\right\}$$
$$¿1,098-1,098$$
$$¿0$$
(3)

2.3
$$\int\frac{(y-1)(y-1)}{y-1}dy$$
$$\int(y-1)dy$$
$$\frac{y^2}{2}-y+c$$
(3)
[20]

QUESTION 3

3.1
$$x+y+10=24.....(1)$$
$$x^2+y^2=100.....(2)$$
$$\ln(1)x=14-y$$
$$\ln(2)(4-y)^2+y^2=100$$
$$y^2-14y+48=0$$
$$(y-8)(y-6)=6$$
$$y_1=8\ cm \quad and \quad y_2=6\ cm$$
$$y \qquad\qquad 2 \quad in \quad (1)$$
$$x_1=6\ cm$$
$$x_2=8\ cm$$
(4)

3.2 $(2x-5)\ln 0,8 = x\ln 2,5$

$x - 0,4463x + 1,116x = 0,916x$

$-1,363x = -1,116$

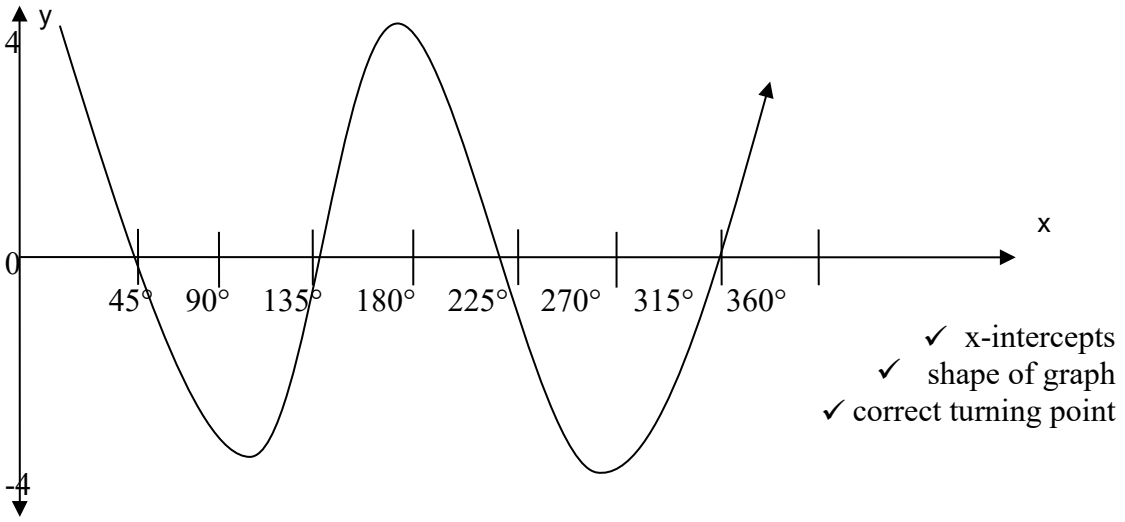
$x = 0,819$ (3)

3.3 $\frac{2RXYZ}{QS} = B^{2P}$

$2P\log B = \log\left(\frac{2RXYZ}{QS}\right)$

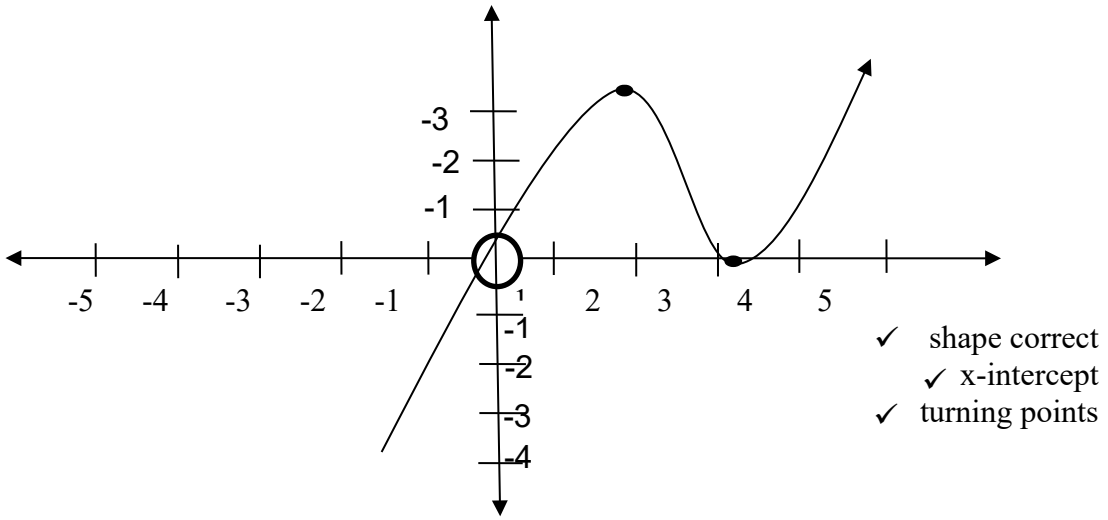
$P = \frac{\log\left(\frac{2RXYZ}{QS}\right)}{2\log B}$ (3)

3.4



(3)

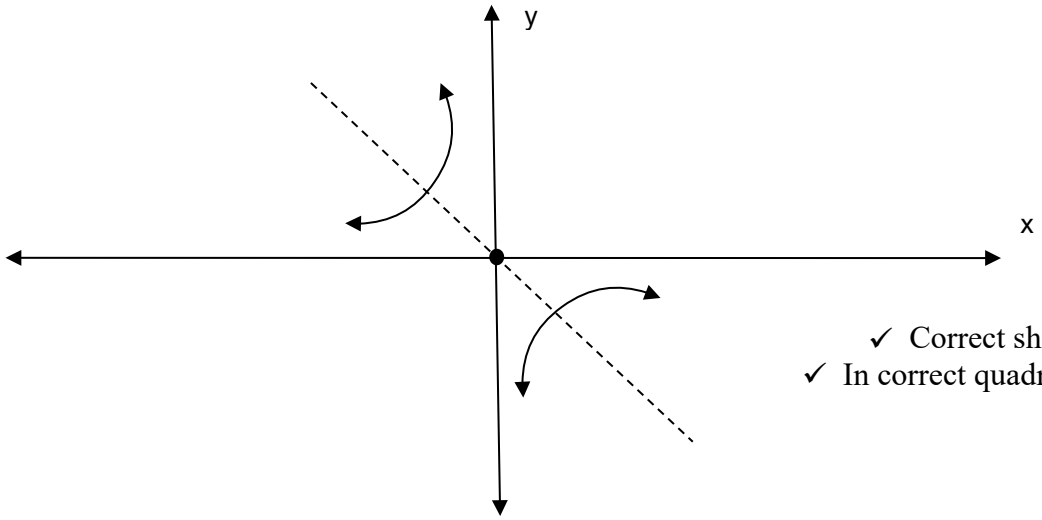
3.5 3.5.1



(3)

3.5.2 $\{x : 0 \leq x \leq 3\, xER\}$ (1)

3.5.3 Continuous (1)



✓ Correct shapes
✓ In correct quadrants

(2)
[20]

QUESTION 4

4.1

$$x-jy=(j-4)+\frac{2+j}{2-j}\times\frac{2+j}{2+j}$$
$$x-jy=(j-4)+\frac{3-j4}{5}$$
$$x-jy=\frac{j5-20+3-j4}{5}$$
$$5(x-jy)=-17+j\checkmark$$
$$5x=-17\ldots\ldots\ldots\ldots\ldots(1)$$
$$x=\frac{-17}{5}\qquad\checkmark$$
$$-5y=1\ldots\ldots\ldots\ldots\ldots\ldots(2)$$
$$y=\frac{-1}{5}\qquad\checkmark$$
$$x-jy=(j-4)+\frac{2+j}{2-j}\times\frac{2+j}{2+j}$$
$$x-jy=(j-4)+\frac{3-j4}{5}$$
$$x-jy=\frac{j5-20+3-j4}{5}$$
$$5(x-jy)=-17+j$$
$$5x=-17\ldots\ldots\ldots\ldots\ldots(1)$$
$$x=\frac{-17}{5}$$
$$-5y=1\ldots\ldots\ldots\ldots\ldots\ldots(2)$$
$$y=\frac{-9}{5}$$

(4)

4.2

$$Z_1=(8,246|\underline{14,036°})^{\frac{-1}{2}}\checkmark$$
$$¿0,348|\underline{7,018°}\qquad\checkmark$$
$$Z_2=3,163|339.517\qquad\checkmark$$
$$Z_1=0,348|\underline{-7,018°}\qquad\checkmark$$
$$¿9,089|\underline{346.535}\qquad\checkmark$$

(5)

4.3 (1)

1

4.4 4.4.1

$$|D|=\begin{vmatrix}2 & 1 & -3\\ 1 & -2 & 5\\ 3 & 4 & -1\end{vmatrix}$$

$$|D|=2\begin{vmatrix}-2 & 5\\ 4 & -1\end{vmatrix}-1\begin{vmatrix}1 & 5\\ 3 & -1\end{vmatrix}-3\begin{vmatrix}1 & -2\\ 3 & 4\end{vmatrix}$$

$$=2(-18)-1(-16)-3(10)$$

$$=-50$$

$$|Dc|=\begin{vmatrix}2 & 1 & 4\\ 1 & -2 & 7\\ 3 & 4 & 8\end{vmatrix}$$

$$=2\begin{vmatrix}-2 & 7\\ 4 & 8\end{vmatrix}-1\begin{vmatrix}1 & 7\\ 3 & 8\end{vmatrix}+4\begin{vmatrix}1 & -2\\ 3 & 4\end{vmatrix}$$

$$=2(-44)-1(-13)+4(10)$$

$$=-35$$

$$c=\frac{|Dc|}{|D|}=\frac{-35}{-50}$$

$$=0,7$$ (6)

4.4.2

$$\begin{vmatrix}2 & -3\\ 3 & -1\end{vmatrix}$$

Minor of -2: $=7$ (2)

4.4.3

$$(+)\begin{vmatrix}1 & -3\\ -2 & 5\end{vmatrix}$$

Co-factor of 3: $=-1$ (2)

[20]

QUESTION 5

5.1
$$3\left(1+\tan^2 y\right)-4\tan y-2=0$$
$$3\tan^2 y-4\tan y+1=0$$
$$(3\tan y-1)(\tan y-1)=0$$
$$\tan y=\frac{1}{3} \quad and \quad \tan y=1$$
$$y_1=18,435° \quad y_3=45°$$
$$y_2=198,434° \quad y_4=225°$$
(6)

5.2
$$RHS=\frac{-\sin x.-\cos x+\cos x}{\left[0-(-1)\right],\cos x}$$
$$=\frac{\sin x\cos x+\cos x}{1.\cos x}$$
$$=\frac{\sin x\cos x}{\cos x}+\frac{\cos x}{\cos x}$$
$$=1+\sin x$$
$$RHS=LHS$$
(4)

5.3
$$=\frac{1-\left(1-2\sin^2\theta\right)}{\tan\theta}$$
$$¿\frac{2\sin\theta}{1}\times\frac{\cos\theta}{\sin\theta}$$
$$¿2\sin\theta\cos\theta$$
$$¿\sin 2\theta$$
(3)

5.4
$$\cos\frac{45°}{2}=\pm\sqrt{\frac{\cos 45+1}{2}}$$
$$=\pm\sqrt{\frac{\frac{\sqrt{2}}{2}+\frac{2}{2}}{2}}$$
$$=\pm\sqrt{\frac{\sqrt{2}+2}{4}}$$
$$=\pm\sqrt{\frac{\sqrt{2}+2}{2}}\ or\ \frac{\sqrt{2}+2}{4}$$
(4)

5.5 $\sin(x-y)=\sin x\cos y-\sin y\cos x$

$=\frac{3}{5}.\frac{8}{9}-\frac{\sqrt{17}}{9}.\frac{4}{5}$ ✓✓

$=\frac{24}{45}-\frac{4\sqrt{17}}{45}$

$=\frac{24-4\sqrt{17}}{45}$

(3)
[20]

100
TOTAL:

Copyright reserved